GATE ME 90 Days Challenge

Day 23/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Machine Design	Static & Fatigue Design	85 days to GATE

How to use this set: 15 questions (5 MCQ, 10 NAT) on design for static loading with the theories of failure (Rankine, Tresca, von Mises), stress concentration and notch sensitivity, fluctuating stresses, the endurance limit and its modifying factors, the S-N diagram, Soderberg, Goodman and Gerber criteria, and Miner's rule for cumulative damage. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

In the σ1–σ2 plane, the safe (non-yielding) region predicted by the maximum shear stress (Tresca) theory for plane stress is bounded by

(A) a hexagon
(B) a square
(C) a circle
(D) an ellipse

Q2 NAT

A point on the surface of a shaft has a bending stress of 80 MPa and a torsional shear stress of 60 MPa. The material has S_{yt} = 300 MPa. The factor of safety by the von Mises theory (round off to two decimal places) is _____

Q3 NAT

A cast-iron bracket (brittle, S_{ut} = 200 MPa) has, at its critical point, a normal stress of 40 MPa and a shear stress of 30 MPa. The factor of safety by the maximum principal stress theory (round off to two decimal places) is _____

Q4 MCQ

For a ductile material in pure shear, the ratio of the shear yield strength predicted by the maximum shear stress theory to that predicted by the von Mises theory is

(A) 0.707
(B) 0.866
(C) 0.5
(D) 1.155

Q5 NAT

A solid shaft transmits a torque of 1000 N·m and carries a bending moment of 800 N·m. It is made of steel with S_{yt} = 320 MPa. Using the von Mises theory and a factor of safety of 2, the minimum diameter (in mm, round off to one decimal place) is _____

Q6 NAT

A notched specimen has a theoretical stress concentration factor of 2.5. Fatigue tests show that its endurance limit is 1/2.05 of that of an unnotched specimen. The notch sensitivity of the material (round off to two decimal places) is _____

Q7 MCQ

Which of the following treatments **increases** the fatigue strength of a steel spring?

(A) Decarburisation of the surface during heat treatment

(B) Shot peening the surface

(C) Rough machining of the surface

(D) Electroplating with chromium

Q8 NAT

A steel component has S_{ut} = 600 MPa. Surface factor = 0.78, size factor = 0.85, reliability factor = 0.868 (95 %), and the fatigue stress concentration factor is 1.6 (applied to the endurance limit). Taking $S_e' = 0.5\ S_{ut}$, the corrected endurance limit (in MPa, round off to one decimal place) is _____

Q9 NAT

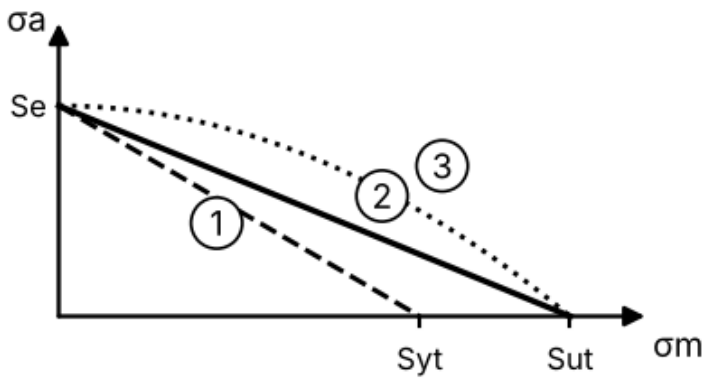
A 20 mm diameter rod is under an axial tensile load that varies between 10 kN and 40 kN. Its corrected endurance limit is 180 MPa and S_{ut} = 540 MPa. The factor of safety by the Goodman criterion (round off to two decimal places) is _____

Q10 NAT

A tie rod carries a steady axial load of 30 kN plus a completely reversed axial load of ±20 kN. The corrected endurance limit is 200 MPa and S_{yt} = 350 MPa. Using the Soderberg criterion with a factor of safety of 2, the minimum rod diameter (in mm, round off to two decimal places) is _____

Q11 MCQ

The figure shows three fatigue failure lines on the σ_m–σ_a diagram. Line 2 (straight, from S_e to S_{ut}) is the



(A) Goodman line

(B) Gerber parabola

(C) Langer (first-cycle yield) line

(D) Soderberg line

Q12 NAT

A steel has S_{ut} = 900 MPa and a corrected endurance limit of 400 MPa at 10^6 cycles. Assuming a straight S-N line on log-log axes from 0.9 S_{ut} at 10^3 cycles to S_e at 10^6 cycles, the fatigue strength for a life of 10^5 cycles (in MPa, round off to one decimal place) is _____

Q13 NAT

A crane hook sees a repeating block of loading: 100 cycles at stress σ1 (life 5×10^4 cycles), 1000 cycles at σ2 (life 5×10^5 cycles) and 10 000 cycles at σ3, which is below the endurance limit. By Miner's rule, the number of blocks to failure is _____

Q14 MCQ

A stress that varies from zero to a maximum value σ_{max} and back to zero (repeated stress, stress ratio R = 0) has

(A) $\sigma_m = 0, \sigma_a = \sigma_{max}$

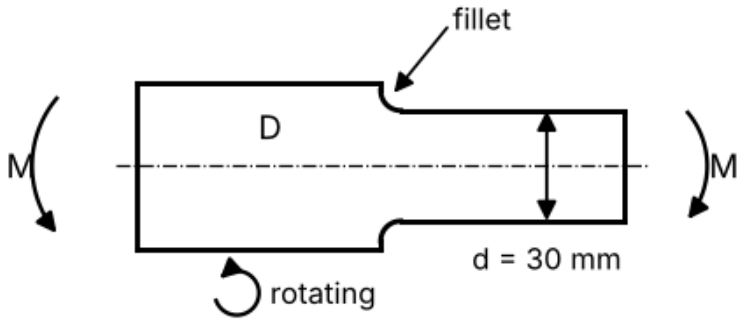
(B) $\sigma_m = \sigma_{max}/2, \sigma_a = \sigma_{max}$

(C) $\sigma_m = \sigma_{max}, \sigma_a = 0$

(D) $\sigma_m = \sigma_a = \sigma_{max}/2$

Q15 NAT

The rotating stepped shaft shown has a smaller diameter of 30 mm at the fillet. For the fillet, $K_t = 1.8$ and q = 0.9. The corrected endurance limit of the unnotched shaft material is 250 MPa. For a factor of safety of 2, the largest bending moment the shaft can carry at the fillet (in N·m, round off to one decimal place) is _____



Answer Key

Q	Type	Answer
Q1	MCQ	(A) a hexagon
Q2	NAT	2.29 (accept 2.27 to 2.30)
Q3	NAT	3.57 (accept 3.55 to 3.59)
Q4	MCQ	(B) 0.866
Q5	NAT	42.2 (accept 41.9 to 42.4)
Q6	NAT	0.7
Q7	MCQ	(B) Shot peening the surface
Q8	NAT	107.9 (accept 107.3 to 108.6)
Q9	NAT	2.42 (accept 2.41 to 2.44)
Q10	NAT	21.75 (accept 21.62 to 21.88)
Q11	MCQ	(A) Goodman line
Q12	NAT	506.1 (accept 503.0 to 509.1)
Q13	NAT	250
Q14	MCQ	(D) $\sigma_m = \sigma_a = \sigma_{max}/2$
Q15	NAT	192.6 (accept 191.5 to 193.8)

Detailed Solutions

Q1 **Answer: (A) a hexagon**

Concept: Tresca: yielding when the largest of $|\sigma1|$, $|\sigma2|$ and $|\sigma1 - \sigma2|$ reaches S_{yt}.

In the first and third quadrants the limit is $|\sigma1| = S_{yt}$ or $|\sigma2| = S_{yt}$ (square edges); in the second and fourth quadrants it is $|\sigma1 - \sigma2| = S_{yt}$ (inclined lines). Together they form a **hexagon**.

The von Mises locus $\sigma1^2 - \sigma1\sigma2 + \sigma2^2 = S_{yt}^2$ is an **ellipse** that passes through all six corners of the hexagon; Rankine gives a **square**.

Exam tip: The largest gap between Tresca and von Mises is in pure shear ($\sigma1 = -\sigma2$): 0.5 S_{yt} versus 0.577 S_{yt}.

Q2 **Answer: 2.29 (accept 2.27 to 2.30)**

Concept: For $\sigma_y = 0$, the von Mises stress is $\sigma_e = \sqrt{(\sigma^2 + 3\tau^2)}$.

$\sigma_e = \sqrt{(80^2 + 3 \times 60^2)} = \sqrt{17200} =$ **131.15 MPa**

$N = S_{yt}/\sigma_e = 300/131.15 =$ **2.29**

Exam tip: Tresca would use $\sqrt{(\sigma^2 + 4\tau^2)} = 144.2$ MPa, giving N = 2.08. Read carefully which theory the question asks for.

Q3 **Answer: 3.57 (accept 3.55 to 3.59)**

Concept: Brittle materials fail when the largest principal stress reaches S_{ut} (Rankine theory).

$\sigma_1 = \sigma/2 + \sqrt{((\sigma/2)^2 + \tau^2)} = 20 + \sqrt{(20^2 + 30^2)} = 20 + 36.06 =$ **56.06 MPa**

$N = 200/56.06 =$ **3.57**

Exam tip: Use Rankine (or Coulomb-Mohr) for brittle materials and von Mises/Tresca for ductile ones. GATE often tests just this choice.

Q4 **Answer: (B) 0.866**

Concept: Tresca: $S_{sy} = 0.5\ S_{yt}$. von Mises: $S_{sy} = S_{yt}/\sqrt{3} = 0.577\ S_{yt}$.

Ratio $= 0.5/0.577 = \sqrt{3}/2 =$ **0.866**

Exam tip: So Tresca is up to about 13 % conservative compared with von Mises; 1.155 is the inverse ratio.

Q5 **Answer: 42.2 (accept 41.9 to 42.4)**

Concept: $\sigma = 32M/\pi d^3$, $\tau = 16T/\pi d^3$; von Mises: $\sqrt{(\sigma^2 + 3\tau^2)} = (32/\pi d^3)\sqrt{(M^2 + \frac{3}{4}T^2)} = S_{yt}/N$.

Equivalent moment $\sqrt{(M^2 + 0.75T^2)} = \sqrt{(800^2 + 0.75 \times 1000^2)} =$ **1179.0 N·m**

Allowable stress $= 320/2 = 160$ MPa

$d^3 = 32 \times 1178983/(\pi \times 160) = 75056\ mm^3 \rightarrow d =$ **42.2 mm**

Exam tip: Tresca uses $\sqrt{(M^2 + T^2)}$ instead of $\sqrt{(M^2 + 0.75T^2)}$, so it always gives a slightly larger shaft.

Q6 **Answer: 0.7**

Concept: K_f = (endurance limit unnotched)/(endurance limit notched) and $K_f = 1 + q(K_t - 1)$.

$K_f =$ **2.05**

$q = (K_f - 1)/(K_t - 1) = (2.05 - 1)/(2.5 - 1) =$ **0.70**

Exam tip: q lies between 0 (no notch effect) and 1 (full theoretical effect). It rises with the notch radius and with material strength.

Q7 **Answer: (B) Shot peening the surface**

Concept: Fatigue cracks start at the surface under tensile stress.

Shot peening leaves a layer of compressive residual stress that delays crack initiation and raises fatigue strength.

Chromium plating leaves tensile residual stress and micro-cracks; decarburisation softens the surface; rough machining adds stress raisers. All three lower it.

Exam tip: Other beneficial treatments: surface rolling, nitriding, case carburising, polishing.

Q8 **Answer: 107.9 (accept 107.3 to 108.6)**

Concept: Marin equation $S_e = k_a\ k_b\ k_c\ k_d\ S_e'$, where the notch factor is $k_d = 1/K_f$.

$S_e' = 0.5 \times 600 =$ **300 MPa**

$S_e = 0.78 \times 0.85 \times 0.868 \times (1/1.6) \times 300 =$ **107.9 MPa**

Exam tip: Apply K_f either to the endurance limit or to the alternating stress, never to both.

Q9 **Answer: 2.42 (accept 2.41 to 2.44)**

Concept: Goodman: $\sigma_a/S_e + \sigma_m/S_{ut} = 1/N$.

$A = \pi/4 \times 20^2 = 314.16$ mm². $P_m = 25$ kN, $P_a = 15$ kN

σ_m = **79.58 MPa**, σ_a = **47.75 MPa**

$1/N = 47.75/180 + 79.58/540 = 0.2653 + 0.1474 = 0.4126 \rightarrow N =$ **2.42**

Exam tip: Always split the load into mean and amplitude first; σ_{max} alone tells you nothing in fatigue.

Q10 **Answer: 21.75 (accept 21.62 to 21.88)**

Concept: Soderberg: $\sigma_a/S_e + \sigma_m/S_{yt} = 1/N$ with $\sigma = P/A$.

$A = N(P_a/S_e + P_m/S_{yt}) = 2 \times (20000/200 + 30000/350) = 2 \times (100.0 + 85.71) =$ **371.43 mm²**

$d = \sqrt{(4A/\pi)} =$ **21.75 mm**

Exam tip: Soderberg is the usual choice when the question gives S_{yt} and asks for a 'safe' design; Goodman needs S_{ut}.

Q11 **Answer: (A) Goodman line**

Concept: All three start at S_e on the σ_a axis; they differ in where and how they meet the σ_m axis. Line 1 (straight, ends at S_{yt}) = **Soderberg**; Line 2 (straight, ends at S_{ut}) = **Goodman**; Curve 3 (parabola to S_{ut}) = **Gerber**.

Exam tip: Langer's line runs from S_{yt} on one axis to S_{yt} on the other and checks yielding in the first cycle.

Q12 **Answer: 506.1 (accept 503.0 to 509.1)**

Concept: On log-log axes, log S falls linearly with log N between $(10^3, 0.9S_{ut})$ and $(10^6, S_e)$.

$0.9\ S_{ut} =$ **810 MPa**. From 10^3 to 10^5 is 2 of the 3 decades.

$S = 810 \times (400/810)^{2/3} = 810 \times 0.6248 =$ **506.1 MPa**

Exam tip: Check your answer lies between S_e and $0.9S_{ut}$, and closer to S_e the larger N is.

Q13 **Answer: 250**

Concept: Miner's rule: failure when $\Sigma\ n_i/N_i = 1$. Stresses below the endurance limit cause no damage ($N = \infty$).

Damage per block $= 100/(5 \times 10^4) + 1000/(5 \times 10^5) + 0 = 0.002 + 0.002 =$ **0.004**

Blocks to failure $= 1/0.004 =$ **250**

Exam tip: The order in which the stress levels are applied does not matter in Miner's rule; that is its main limitation.

Q14 **Answer: (D) $\sigma_m = \sigma_a = \sigma_{max}/2$**

Concept: $\sigma_m = (\sigma_{max} + \sigma_{min})/2$ and $\sigma_a = (\sigma_{max} - \sigma_{min})/2$.

With $\sigma_{min} = 0$: $\sigma_m = \sigma_a =$ **$\sigma_{max}/2$**.

Exam tip: $R = -1$ is completely reversed ($\sigma_m = 0$), $R = 0$ is repeated, $R = 1$ is static.

Q15 **Answer: 192.6 (accept 191.5 to 193.8)**

Concept: A rotating shaft under a steady bending moment sees a completely reversed bending stress, so $\sigma_a = 32M/\pi d^3$ and $\sigma_m = 0$.

$K_f = 1 + 0.9(1.8 - 1) = \mathbf{1.72}$

Allowable nominal stress $= S_e/(K_f N) = 250/(1.72 \times 2) = \mathbf{72.67\ MPa}$

$M = \sigma \pi d^3/32 = 72.67 \times \pi \times 30^3/32 = \mathbf{192.6\ N{\cdot}m}$

Exam tip: Use the smaller diameter at the fillet, because that is where the nominal stress is calculated for K_t.

What next?

Day 23 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

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Next, Day 24/90: Weekly Revision, Week 4 Revision. Comment **GATE90** on our reel to get all links.
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